Nonparametric Spectral Tests for Stationarity in Nonlinear Time Series (WIP)

Peter Eremeev

Advisor: Dr. Danna Zhang

Department of Mathematics

May 5th, 2026 *Preliminary Draft*

1 Introduction

Stationarity is a foundational assumption in time-series inference. Consistency of spectral density estimators, the asymptotic distribution of M-estimators, and the localization of change points are all derived under the assumption that the second-order structure of the process does not vary with time. Testing whether this assumption holds is therefore a precondition for any downstream procedure that takes it as given.

Tests of second-order stationarity have been largely studied (see Priestley and Rao (1969), Dette and Paparoditis (2009), and Dwivedi and Subba Rao (2011)); the cleanest formulation in this literature is the L^2 minimum-distance functional of Dette et al. (2011),

$$D^2 \;=\; \min_g \int_{-\pi}^{\pi} \int_0^1 \big(f(u,\lambda) - g(\lambda)\big)^2 \, du \, d\lambda,$$

where $f(u,\lambda)$ is the spectral density of the locally stationary process and the minimum is calculated over the set of all spectral densities g for stationary processes. Note that $D^2 = 0$ if and only if there exists a function $f : [-\pi,\pi] \to \mathbb{C}$ such that

$$H_0 : f(u,\lambda) = f(\lambda) \quad \text{a.e. on } [0,1] \times [-\pi,\pi]$$

is satisfied. The asymptotic theory for D^2 in Dette et al. (2011), however, is established under conditions that hold for linear or gaussian processes, and fails for nonlinear recursive mechanisms such as ARCH, GARCH, threshold autoregressive, and bilinear models which are of interest for modeling of economic time series.

In this work, we develop a test that (i) avoids parametric specification of the data generating process (ii) admits nonlinear dependence at moment orders attainable by ARCH-type processes, and (iii) admits smooth time-variation in the spectral density under the alternative. Under these conditions we establish a joint central limit theorem for $(\hat{D}_{1,T}, \hat{D}_{2,T})$ and consequently for $\hat{D}_T^2$, derive the limiting variance in closed form, and obtain a consistent estimator $\hat{\tau}_0^2$ under H_0 with an explicit bias correction.

The thesis is organized as follows. Section 2 introduces the nonparametric locally stationary framework, the functional dependence measure, and the geometric moment contracting condition. Section 3 constructs the test statistic from local periodogram ordinates, establishes the joint central limit theorem for $(\hat{D}_{1,T}, \hat{D}_{2,T})$, derives the asymptotic distribution of $\hat{D}_T^2$ under H_0 and the alternative, and exhibits the consistent variance estimator that calibrates the rejection rule. Section 4 reports finite-sample properties on time-varying bilinear and ARCH(1) processes and applies the procedure to a financial time series for which existing piecewise formulations are inadequate.

2 Nonparametric Locally Stationary Process

Parametric locally stationary processes with time-varying coefficients have been largely studied; see, for example, time-varying AR models (Subba Rao (1970), Dahlhaus (1997), Moulines et al. (2005)), ARMA models (Grenier (1983), Dahlhaus and Polonik (2009)), ARCH and GARCH models (Dahlhaus and Subba Rao (2006, 2007), Hafner and Linton (2010), Fryzlewicz and Subba Rao (2011)). In this work, we consider nonparametric locally stationary processes. Let $(X_{t,T})_{t=1}^{T}$ be the observed sequence generated from the model

$$X_{t,T} = X_t(t/T) = h(t/T, \mathcal{F}_t), \tag{1}$$

where $\mathcal{F}_t = (\ldots, \varepsilon_{t-1}, \varepsilon_t)$, ε_t, $t \in \mathbb{Z}$ are i.i.d. random variables, $h(\cdot,\cdot)$ is a measurable function such that the process $X_t(u) = h(u, \mathcal{F}_t)$ is well defined.

Assumption 2.1. Assume that the uniform stochastic Lipschitz continuity holds: there exists some constant $\mathcal{C} > 0$ such that

$$\|h(u, \mathcal{F}_t) - h(v, \mathcal{F}_t)\|_2 \leq \mathcal{C}|u - v|, \text{ for all } u, v \in [0, 1], \tag{2}$$

where $\|X\|_2 = (\mathbb{E}X^2)^{1/2}$ for a random variable X.

The stochastic continuity condition (2) indicates that $X_t(u) = h(u, \mathcal{F}_t)$ changes smoothly with respect to u. One has local stationarity in the sense that the sequence $(X_{t,T})$ can be approximated by the stationary process $X_{tj}(u) = h(u, \mathcal{F}_t)$ with a fixed u in view of

$$\|X_{t,T} - X_t(u)\|_2 \leq \mathcal{C}\Big|\frac{t}{T} - u\Big|$$

which converges to 0 if $t/T - u \to 0$. For convenience of notation, we shall abbreviate $X_{t,T}$ as X_t.

Zhou (2010) performed nonparametric specification tests of quantile curves under the framework (1). In the stationary case where $h(u, \cdot)$ does not depend on u, (1) then becomes

$X_t = h(\mathcal{F}_t)$ and one can let $\mathcal{C} = 0$ in (2). The framework $X_t = h(\mathcal{F}_t)$ has a clear physical meaning, where (ε_t) are the inputs and (X_t) are the outputs. Such a representation is very natural for modeling time series. It was studied by Wiener (1958) for representing stationary and ergodic processes, and it is sometimes called nonlinear Wold representation. It is general enough to include a wide range of stochastic processes (Rosenblatt (1971), Priestley (1988), Tong (1990), Wu (2005), Tsay (2005)). It subsumes linear processes, their nonlinear transforms, as well as the Volterra processes that involve interactions between the innovations. The representation $X_t = h(\mathcal{F}_t)$ also includes recursive model of the form $X_t = h(X_{t-1}, \varepsilon_t)$, which includes Markov chain models and nonlinear autoregressive models such as threshold autoregressive models, autoregressive models with conditional heteroscedasticity and exponential autoregressive models, bilinear autoregressive models etc. By allowing the data-generating function h to change flexibly over time $u \in [0,1]$, it extends a large number of existing stationary processes into their non-stationary counterparts in a natural way.

To develop an asymptotic theory for the time-varying spectral estimate, we need to introduce appropriate dependence measures. Assume that $\sup_{u\in[0,1]} \|h(u, \mathcal{F}_0)\|_q < \infty$ for some $q \geq 1$. Let ε'_t, $t \in \mathbb{Z}$ be an i.i.d. copy of ε_t, $t \in \mathbb{Z}$. For $t \geq 0$, we define the functional dependence measure

$$\delta_{t,q} = \sup_{u\in[0,1]} \|h(u, \mathcal{F}_t) - h(u, \mathcal{F}_{t,\{0\}})\|_q, \tag{3}$$

where $\mathcal{F}_{t,\{0\}} = (\ldots, \varepsilon_{-1}, \varepsilon'_0, \varepsilon_1, \ldots, \varepsilon_{t-1}, \varepsilon_t)$ is a coupled version of $\mathcal{F}_t$ with ε_0 in $\mathcal{F}_t$ replaced by ε'_0. Note that $\mathcal{F}_{t,\{0\}} = \mathcal{F}_t$ if $t < 0$. Hence, $\delta_{t,q} = 0$ for $t < 0$. Wu (2005) introduced a functional dependence measure for stationary processes in which the data-generating mechanism h does not vary with time u. In our setting, the quantity $\delta_{t,q}$ measures the dependence of $h(u, \mathcal{F}_t)$ on the single input ε_0 over $u \in [0,1]$, which can be viewed as the uniform dependence measure with lag t for locally stationary processes. Equipped with the dependence measures (3), we define the q-th dependence adjusted moment.

Assumption 2.2. Assume $\mathbb{E}(|X_t|^q) < \infty$ for $q = 8 + \delta$, with $\delta > 0$, and that there exists some constant $\rho \in (0,1)$ such that

$$\|X_{\cdot}\|_q = \sup_{m\geq 0} \rho^{-m} \Delta_{m,q} < \infty, \text{ where } \Delta_{m,q} = \sum_{t=m}^{\infty} \delta_{t,q}. \tag{4}$$

In Assumption (2.2), a geometric moment contracting (GMC) condition is imposed on the cumulative (tail) dependence measure $\Delta_{m,q} = \sum_{t=m}^{\infty} \delta_{t,q}$ for $q = 8 + \delta$ by noting that $\Delta_{m,q} \leq \|X_\cdot\|_q \cdot \rho^m$ for all $m \geq 0$. We can interpret the quantity $\|X_\cdot\|_q$ as the q-th moment of the component process $\{X_t\}$ by taking dependence into account. Elementary calculations show that, if X_t, $t \in \mathbb{Z}$, are i.i.d., then $\|X_t\|_q \leq \|X_\cdot\|_q \leq 2\|X_t\|_q$, suggesting that the dependence adjusted moment is equivalent to the classical $\mathcal{L}^q$ moment for the independent case.

Under the framework (1), the time-varying spectral density matrix function is defined as

$$f(u, \lambda) = \frac{1}{2\pi} \sum_{k \in \mathbb{Z}} \gamma_k(u) \exp(-\iota k \theta), \text{ where } \iota = \sqrt{-1}.$$

where $\gamma_k(u)$ is the time-varying autocovariance matrix with lag k, defined by $\gamma_k(u) = \mathbb{E} X_0(u) X_k(u)$ for $X_t(u) = h(u, \mathcal{F}_t)$.

3 Test of Stationarity

Lemma 3.1 (Lemma 1 in Dette et al. (2011))**.** *The minimal distance defined in (1) can be calculated by*

$$D^2 = \int_{-\pi}^{\pi} \int_0^1 f^2(u, \lambda) du d\lambda - \int_{-\pi}^{\pi} \Big(\int_0^1 f(u, \lambda) du \Big)^2 d\lambda. \tag{5}$$

By Lemma 3.1, we shall estimate the two integrals

$$D_1 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_0^1 f^2(u, \lambda) du d\lambda, \tag{6}$$

$$D_2 = \frac{1}{4\pi} \int_{-\pi}^{\pi} \Big(\int_0^1 f(u, \lambda) du \Big)^2 d\lambda. \tag{7}$$

Without loss of generality, assume that $T = NM$, where M and N are positive integers, with M the number of blocks and N an even block size. We define the local periodogram around time u based on N observations by

$$I_N(u, \lambda) = \frac{1}{2\pi N} \Big| \sum_{t=1}^{N} X_{\lfloor uT \rfloor - N/2 + t} \exp(-\iota \lambda t) \Big|^2. \tag{8}$$

And $X_t = 0$ if $t \notin \{1, \ldots, T\}$. We estimate the integrals in (6) and (7) by its Riemann approximation in time and frequency. Define $u_j = \frac{N(j-1)+N/2}{T}$ as the mid-point of each block and $\lambda_k = \frac{2\pi k}{N}$ as the Fourier frequency. Let

$$\hat{D}_{1,T} = \frac{1}{T} \sum_{k=1}^{N/2} \sum_{j=1}^{M} I_N(u_j, \lambda_k)^2, \tag{9}$$

$$\hat{D}_{2,T} = \frac{1}{N} \sum_{k=1}^{N/2} \Big(\frac{1}{M} \sum_{j=1}^{M} I_N(u_j, \lambda_k) \Big)^2 \tag{10}$$

Then the minimal distance can be estimated by

$$\hat{D}_T^2 = 2\pi \hat{D}_{1,T} - 4\pi \hat{D}_{2,T}. \tag{11}$$

Assumption 3.1. Assume that $N \to \infty,\ M \to \infty,\ N/T \to 0$.

Theorem 3.2. *Under Assumptions 2.1, 2.2 and 3.1, we have*

$$\sqrt{T}[(\hat{D}_{1,T}, \hat{D}_{2,T})^\top - (D_1, D_2 + d_{N,T})^\top] \xrightarrow{d} N(0, \Sigma), \tag{12}$$

where

$$\Sigma = \begin{pmatrix} \frac{5}{\pi} \int_{-\pi}^{\pi} \int_0^1 f^4(u,\lambda) du d\lambda & \frac{2}{\pi} \int_{-\pi}^{\pi} (\int_0^1 f(u,\lambda) du \int_0^1 f^3(u,\lambda) du) d\lambda \\ \frac{2}{\pi} \int_{-\pi}^{\pi} (\int_0^1 f(u,\lambda) du \int_0^1 f^3(u,\lambda) du) d\lambda & \frac{1}{\pi} \int_{-\pi}^{\pi} [(\int_0^1 f(u,\lambda) du)^2 \int_0^1 f^2(u,\lambda) du] d\lambda \end{pmatrix}$$

and

$$d_{N,T} = \frac{N}{4\pi T} \int_{-\pi}^{\pi} \int_0^1 f^2(u,\lambda) du d\lambda.$$

Proof. Assuming Assumption 2.2 and applying similar arguments of proving Corollary 2.1 in Shao and Wu (2007), we can write

$$I_N(u_j, \lambda_k) = f(u_j, \lambda_k) E_{j,k},$$

and obtain that $E_{j,k}$ are asymptotically i.i.d. standard exponential random variables [Exp(1)]. For locally stationary processes under Assumption 2.1 and 2.2, by Corollary 4.3 in Zhang and Wu (2021), we have

$$\mathbb{E}[I_N(u_j, \lambda_k)] = f(u_j, \lambda_k) + O(N^{-1}).$$

(i) Mean and variance of $\hat{D}_{1,T}$.

For notational simplicity, denote

$$I_{jk} := I_N(u_j, \lambda_k) \text{ and } f_{jk} := f(u_j, \lambda_k).$$

By the asymptotic exponential distribution,

$$\mathbb{E}(I_{jk}^2) = 2f_{jk}^2 + o(1).$$

Therefore,

$$\mathbb{E}(\hat{D}_{1,T}) = \frac{1}{T}\sum_{j=1}^{M}\sum_{k=1}^{N/2} 2f_{jk}^2 + o(1).$$

Using the Riemann approximation,

$$\frac{1}{T}\sum_{j=1}^{M}\sum_{k=1}^{N/2} f_{jk}^2 \to \frac{1}{2\pi}\int_0^1 \int_0^\pi f^2(u,\lambda)\, d\lambda\, du.$$

Since $f(u, -\lambda) = f(u, \lambda)$,

$$\mathbb{E}(\hat{D}_{1,T}) \to \frac{1}{2\pi}\int_{-\pi}^{\pi}\int_0^1 f^2(u,\lambda)\, du\, d\lambda = D_1.$$

Hence $\hat{D}_{1,T}$ is asymptotically unbiased. Define $Z_{jk} = I_{jk}^2 - \mathbb{E}(I_{jk}^2)$. Then

$$\sqrt{T}(\hat{D}_{1,T} - D_1) = \frac{1}{\sqrt{T}}\sum_{j=1}^{M}\sum_{k=1}^{N/2} Z_{jk} + o_p(1).$$

It suffices to study the variance of the double sum. Since $\frac{I_N(u_j,\lambda_k)}{f(u_j,\lambda_k)} = E_{j,k} \xrightarrow{d} \exp(1)$ uniformly in j, k,

$$\sup_{j,k} |\mathbb{E}(E_{j,k}^r) - \mathbb{E}(E^r)| \to 0, \text{ for } r = 2, 4$$

where $E \sim \mathrm{Exp}(1)$. Since $\mathbb{E}(E^2) = 2$ and $\mathbb{E}(E^4) = 24$, we obtain $\mathrm{Var}(E^2) = 24 - 2^2 = 20$. Therefore,

$$\mathbb{E}(I_{jk}^2) = 2f_{jk}^2 + o(1),$$
$$\mathrm{Var}(I_{jk}^2) = 20f_{jk}^4 + o(1).$$

Moreover, for distinct frequencies and blocks,

$$\mathrm{Cov}(I_{jk}^2, I_{j'\ell}^2) = o(1),$$

whenever $(j,k) \neq (j',\ell)$. Hence the leading variance contribution comes from the diagonal terms only. Thus,

$$\mathrm{Var}(\hat{D}_{1,T}) = \frac{1}{T^2}\sum_{j=1}^{M}\sum_{k=1}^{N/2} 20 f_{jk}^4 + o(T^{-1}).$$

Multiplying by T,

$$T\mathrm{Var}(\hat{D}_{1,T}) = \frac{20}{T}\sum_{j=1}^{M}\sum_{k=1}^{N/2} f_{jk}^4 + o(1).$$

Applying again the Riemann approximation,

$$\frac{1}{T}\sum_{j=1}^{M}\sum_{k=1}^{N/2} f_{jk}^4 \to \frac{1}{4\pi}\int_{-\pi}^{\pi}\int_0^1 f^4(u,\lambda)\,du\,d\lambda.$$

Therefore,

$$T\mathrm{Var}(\hat{D}_{1,T}) \to \frac{5}{\pi}\int_{-\pi}^{\pi}\int_0^1 f^4(u,\lambda)\,du\,d\lambda.$$

(ii) Mean and variance of $\hat{D}_{2,T}$.

Define $\bar{I}_k = \frac{1}{M}\sum_{j=1}^{M} I_{jk}$. Then $\hat{D}_{2,T} = \frac{1}{N}\sum_{k=1}^{N/2} \bar{I}_k^2$. Define $\mu_k := \frac{1}{M}\sum_{j=1}^{M} f_{jk}$. Using the decomposition $I_{jk} = f_{jk} + (I_{jk} - f_{jk})$, we can obtain

$$\bar{I}_k = \mu_k + \frac{1}{M}\sum_{j=1}^{M}(I_{jk} - f_{jk}).$$

Hence

$$\bar{I}_k^2 = \Big(\mu_k + \frac{1}{M}\sum_{j=1}^{M}(I_{jk} - f_{jk})\Big)^2 = \mu_k^2 + \frac{2\mu_k}{M}\sum_{j=1}^{M}(I_{jk} - f_{jk}) + \Big(\frac{1}{M}\sum_{j=1}^{M}(I_{jk} - f_{jk})\Big)^2.$$

Therefore,

$$\hat{D}_{2,T} = \frac{1}{N}\sum_{k=1}^{N/2}\mu_k^2 + \frac{2}{NM}\sum_{k=1}^{N/2}\sum_{j=1}^{M}\mu_k(I_{jk} - f_{jk}) + \frac{1}{N}\sum_{k=1}^{N/2}\Big(\frac{1}{M}\sum_{j=1}^{M}(I_{jk} - f_{jk})\Big)^2.$$

We now compute the expectation of the last term. Since

$$\mathbb{E}(I_{jk} - f_{jk}) = o(1),$$

and covariances between different blocks are asymptotically negligible,

$$\mathbb{E}\Big[\Big(\frac{1}{M}\sum_{j=1}^{M}(I_{jk}-f_{jk})\Big)^2\Big]=\frac{1}{M^2}\sum_{j=1}^{M}\mathrm{Var}(I_{jk})+o(M^{-1}).$$

Using $\mathrm{Var}(I_{jk})=f_{jk}^2+o(1)$, we can obtain

$$\mathbb{E}\Big[\Big(\frac{1}{M}\sum_{j=1}^{M}(I_{jk}-f_{jk})\Big)^2\Big]=\frac{1}{M^2}\sum_{j=1}^{M}f_{jk}^2+o(M^{-1}).$$

Consequently,

$$\mathbb{E}(\hat{D}_{2,T})=\frac{1}{N}\sum_{k=1}^{N/2}\mu_k^2+\frac{1}{NM^2}\sum_{k=1}^{N/2}\sum_{j=1}^{M}f_{jk}^2+o\big(\frac{N}{T}\big).$$

The first term converges to

$$D_2=\frac{1}{4\pi}\int_{-\pi}^{\pi}\Big(\int_0^1 f(u,\lambda)\,du\Big)^2 d\lambda.$$

For the second term, since $T=MN$,

$$\frac{1}{NM^2}\sum_{j=1}^{M}\sum_{k=1}^{N/2}f_{jk}^2=d_{N,T}+o\big(\frac{N}{T}\big).$$

Therefore,

$$\mathbb{E}(\hat{D}_{2,T})=D_2+d_{N,T}+o\big(\frac{N}{T}\big).$$

Using the previous expansion,

$$\hat{D}_{2,T}-D_2=\frac{2}{NM}\sum_{k=1}^{N/2}\sum_{j=1}^{M}\mu_k(I_{jk}-f_{jk})+o_p(T^{-1/2}).$$

Hence

$$\sqrt{T}(\hat{D}_{2,T}-D_2-d_{N,T})=\frac{2\sqrt{T}}{NM}\sum_{k=1}^{N/2}\sum_{j=1}^{M}\mu_k(I_{jk}-f_{jk})+o_p(1).$$

Since

$$\mathrm{Var}(I_{jk})=f_{jk}^2+o(1),$$

and covariances are asymptotically negligible,

$$T\mathrm{Var}(\hat{D}_{2,T})=\frac{4T}{N^2M^2}\sum_{k=1}^{N/2}\sum_{j=1}^{M}\mu_k^2f_{jk}^2+o(1).$$

For $T = MN$, $4T/(N^2M^2) = 4/(NM)$. Thus

$$T\mathrm{Var}(\hat{D}_{2,T}) = \frac{4}{NM}\sum_{k=1}^{N/2}\sum_{j=1}^{M}\mu_k^2 f_{jk}^2 + o(1).$$

Applying the Riemann approximation,

$$\frac{1}{NM}\sum_{k=1}^{N/2}\sum_{j=1}^{M}\mu_k^2 f_{jk}^2 \to \frac{1}{4\pi}\int_{-\pi}^{\pi}\Big(\int_0^1 f(u,\lambda)\,du\Big)^2\Big(\int_0^1 f^2(u,\lambda)\,du\Big)d\lambda.$$

Therefore

$$T\mathrm{Var}(\hat{D}_{2,T}) \to \frac{1}{\pi}\int_{-\pi}^{\pi}\Big[\Big(\int_0^1 f(u,\lambda)du\Big)^2\Big(\int_0^1 f^2(u,\lambda)du\Big)\Big]d\lambda.$$

(iii) Covariance between $\hat{D}_{1,T}$ and $\hat{D}_{2,T}$.

Recall that

$$\hat{D}_{1,T} = \frac{1}{T}\sum_{j,k} I_{jk}^2.$$

Using the leading expansion of $\hat{D}_{2,T}$,

$$\hat{D}_{2,T} - D_2 = \frac{2}{NM}\sum_{j,k}\mu_k(I_{jk} - f_{jk}) + o_p(T^{-1/2}).$$

Hence

$$\mathrm{Cov}(\hat{D}_{1,T}, \hat{D}_{2,T}) = \frac{2}{T\,NM}\sum_{j,k}\mu_k \mathrm{Cov}(I_{jk}^2, I_{jk}) + o(T^{-1}).$$

Since $I_{jk} = f_{jk}E_{jk}$ where $E_{jk} \xrightarrow{d} \mathrm{Exp}(1)$,

$$\mathrm{Cov}(I_{jk}^2, I_{jk}) = f_{jk}^3 \mathrm{Cov}(E_{jk}^2, E_{jk}).$$

Since

$$\mathbb{E}(E) = 1, \qquad \mathbb{E}(E^2) = 2, \qquad \mathbb{E}(E^3) = 6,$$

we obtain

$$\mathrm{Cov}(E^2, E) = 6 - 2 = 4.$$

Therefore

$$\mathrm{Cov}(I_{jk}^2, I_{jk}) = 4f_{jk}^3 + o(1).$$

Thus

$$T\mathrm{Cov}(\hat{D}_{1,T}, \hat{D}_{2,T}) = \frac{8}{NM} \sum_{j,k} \mu_k f_{jk}^3 + o(1).$$

Applying the Riemann approximation,

$$\frac{1}{NM} \sum_{j,k} \mu_k f_{jk}^3 \to \frac{1}{4\pi} \int_{-\pi}^{\pi} \Big(\int_0^1 f(u,\lambda)\, du \Big) \Big(\int_0^1 f^3(u,\lambda)\, du \Big) d\lambda.$$

Hence

$$T\mathrm{Cov}(\hat{D}_{1,T}, \hat{D}_{2,T}) \to \frac{2}{\pi} \int_{-\pi}^{\pi} \Big(\int_0^1 f(u,\lambda)\, du \Big) \Big(\int_0^1 f^3(u,\lambda)\, du \Big) d\lambda.$$

By similar arguments as Theorem 21 in Xiao and Wu (2011), the absolute summability of 8th-order cumulants of $\{X_t\}$ holds under the GMC condition in Assumption 2.2. It implies the absolute summability of cumulants of order up to four for the quadratic forms I_{jk}^2 . Since $Z_{jk} = I_{jk}^2 - \mathbb{E}(I_{jk}^2)$ is a 4th-order functional of $\{X_t\}$, its 4th-order cumulant involves only cumulants of $\{X_t\}$ up to order 8. Consequently,

$$\sum_{j,k} \mathrm{cum}_4(Z_{jk}) = o(T).$$

Hence, the asymptotic normality can be established by applying the Lindeberg CLT and the Cramér-Wold device. □

Theorem 3.3. *Under the assumptions of Theorem 3.2, we have*

$$\sqrt{T}(\hat{D}_T^2 - D^2 + 4\pi d_{N,T}) \xrightarrow{d} N(0, \tau^2) \tag{13}$$

where

$$\begin{aligned} \tau^2 \;&=\; 20\pi \int_{-\pi}^{\pi} \int_0^1 f^4(u,\lambda) du d\lambda - 32\pi \int_{-\pi}^{\pi} \Big(\int_0^1 f(u,\lambda) du \int_0^1 f^3(u,\lambda) du \Big) d\lambda \\ &\quad +16\pi \int_{-\pi}^{\pi} \Big[\Big(\int_0^1 f(u,\lambda) du \Big)^2 \int_0^1 f^2(u,\lambda) du \Big] d\lambda. \end{aligned}$$

Proof of Theorem 3.3. Note that $\hat{D}_T^2 = 2\pi \hat{D}_{1,T} - 4\pi \hat{D}_{2,T}$. The asymptotic distribution of $\hat{D}_T^2$ in (13) then follows from Theorem 3.2 and the delta method.

□

The asymptotic bias $4\pi d_{N,T}$ is estimated by

$$B_T = \frac{2\pi N}{T^2} \sum_{k=1}^{N/2} \sum_{j=1}^{M} I_N(u_j, \lambda_k)^2.$$

Lemma 3.4. *Under the assumptions of Theorem 3.2, we have*

$$\sqrt{T}(B_T - 4\pi d_{N,T}) \xrightarrow{p} 0.$$

Proof of Lemma 3.4. Since $B_T = \frac{2\pi N}{T} \hat{D}_{1,T}$, it follows from Theorem 1 that

$$\sqrt{T}(B_T - 4\pi d_{N,T}) = \frac{2\pi N}{T} \sqrt{T}(\hat{D}_{1,T} - D_1) \xrightarrow{p} 0.$$

□

Under the stationarity, i.e. $H_0 : f(u, \lambda) = f(\lambda)$, the asymptotic variance τ^2 can be computed as $\tau_0^2 = 4\pi \int_{-\pi}^{\pi} f^4(\lambda) d\lambda$. We estimate τ_0^2 by

$$\hat{\tau}_0^2 = \frac{2\pi^2}{3T} \sum_{k=1}^{N/2} \sum_{j=1}^{M} I_N(u_j, \lambda_k)^4.$$

Theorem 3.5. *Under the assumptions of Theorem 3.2 and under $H_0 : f(u, \lambda) = f(\lambda)$, we have $\hat{\tau}_0^2 \xrightarrow{p} \tau_0^2$.*

Proof of Theorem 3.5. The proof follows the similar arguments to those in Theorem 3.2 and is therefore omitted. □

The null hypothesis $H_0 : f(u, \lambda) = f(\lambda)$ is rejected if $\hat{D}_T^2 + B_T \geq \frac{\hat{\tau}_0^2}{\sqrt{T}} z_{1-\alpha}$.

4 Numerical Study

4.1 Simulation

The application to data of the extension discussed in Section 2, necessitates the selection of the parameters M, N from the local periodogram (8) when confronted with an empirical finite sample. Higher values of M correspond to a more granular partition over rescaled time $u \in [0, 1]$, implying greater sensitivity to time-variation in $f(u, \lambda)$. However, larger

M implies smaller block size N, so each local periodogram $I_N(u_j, \lambda_k)$ is computed from fewer observations. Since the periodogram is an asymptotically unbiased but inconsistent estimator of the spectral density, the squared deviations that enter $\hat{D}_T$ are noisier at smaller N. The consequence is upward distortion of Type 1 Error with the test rejecting H_0 more often than the nominal level α warrants. Conversely, smaller M allocates more observations to each block, reducing periodogram noise and bringing the empirical rejection rate under H_0 closer to the nominal level. But with fewer blocks, the Riemann approximation over u becomes coarser, smooth departures from stationarity are averaged away within each block, and the test loses power to detect them. We now turn to specifics by discussing two data-generating processes; the Time-Varying Bilinear Model, and Time-Varying ARCH(1) model. The latter is crucial for N, M selection of econometric analysis performed in (4.2).

Time-Varying Bilinear

The bilinear model provides a natural first test case for the extended test, the data-generating function $h(u, \mathcal{F}_t)$ is nonlinear innovations, the process admits neither an AR nor an MA representation, and yet the admissible parameter region is wide enough to accommodate large departures from H_0. This last property makes it possible to evaluate power at amplitudes where the test should produce clear rejections. Hence, we consider the data generating process defined by,

$$X_{t,T} = \beta(t/T)\, X_{t-1}\, Z_{t-1} + Z_t \tag{14}$$

Where $Z_t \overset{\text{iid}}{\sim} N(0, 1)$ and $\beta(u) = a + b\cos(2\pi u)$. Under $b = 0$ the coefficient is constant and the process reduces to the stationary bilinear model $X_t = a\, X_{t-1} Z_{t-1} + Z_t$; under $b > 0$ the coefficient oscillates over $[a - b,\ a + b]$ for $u \in [0, 1]$, inducing a time-varying spectral density. A stability requirement necessitates $|a| + |b| < 1$.

Since this process is recursive, we compute the deviations after feeding a burn-in of 500 observations under the stationary regime ($\beta \equiv a$). The bilinear interaction at $a = 0.1$ is weak, producing a near-Gaussian marginal distribution, while at $a = 0.3$ the marginal distribution exhibits far heavier tails with the periodogram-based variance estimator $\hat{\tau}^2_{H_0}$ converging more slowly. The comparison between the two baseline regimes illustrates a tension inherent to the test: processes with stronger nonlinear dependence generate larger spectral distances D^2 (5) for the same amplitude b, but they also produce greater finite-sample size distortion. The $M = 8$ configuration offers the best compromise, holding null rejection rates within roughly 2–4 percentage points of the nominal 5 percent level across both regimes while retaining meaningful power at moderate and large b.

Table 1: $a = 0.1$ (mild nonlinearity).

			$b = 0$ (H_0)		$b = 0.2$ (H_1)		$b = 0.5$ (H_1)	
T	N	M	5%	10%	5%	10%	5%	10%
256	32	8	0.043	0.111	0.066	0.154	0.293	0.430
256	16	16	0.092	0.194	0.130	0.258	0.459	0.605
512	64	8	0.045	0.103	0.061	0.138	0.355	0.483
512	32	16	0.074	0.162	0.115	0.232	0.546	0.676
1024	128	8	0.044	0.103	0.071	0.147	0.485	0.603
1024	64	16	0.070	0.143	0.104	0.200	0.623	0.735
2048	256	8	0.049	0.105	0.071	0.148	0.605	0.716
2048	128	16	0.063	0.132	0.095	0.196	0.771	0.848
2048	64	32	0.106	0.202	0.195	0.307	0.890	0.935

Table 2: $a = 0.3$ (strong nonlinearity)

			$b = 0$ (H_0)		$b = 0.2$ (H_1)		$b = 0.5$ (H_1)	
T	N	M	5%	10%	5%	10%	5%	10%
256	32	8	0.087	0.181	0.206	0.332	0.676	0.770
256	16	16	0.222	0.363	0.363	0.509	0.807	0.874
512	64	8	0.080	0.164	0.250	0.371	0.845	0.893
512	32	16	0.188	0.312	0.398	0.530	0.924	0.955
1024	128	8	0.073	0.142	0.313	0.440	0.964	0.980
1024	64	16	0.144	0.260	0.449	0.582	0.983	0.991
2048	256	8	0.068	0.144	0.412	0.558	0.997	0.999
2048	128	16	0.115	0.211	0.528	0.659	0.999	1.000
2048	64	32	0.252	0.392	0.708	0.807	1.000	1.000

Time-Varying ARCH(1)

For the purposes of analyzing more complex time series, we consider the canonical model of conditional heteroskedasticity. Note, that its moment structure imposes much tighter constraints on the admissible amplitude of time-variation. Hence we consider the ARCH(1) process defined by,

$$X_{t,T} = \sigma_{t,T}\, Z_t, \qquad \sigma^2_{t,T} = \alpha_0 + \alpha_1(t/T)\, X^2_{t-1,T}, \qquad Z_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{15}$$

with $\alpha_0 > 0$ fixed and $\alpha_1(u) = a + b\cos(2\pi u)$. Under $b = 0$ the process reduces to the stationary ARCH(1) model of Engle (1982); under $b > 0$ the ARCH coefficient varies smoothly with rescaled time, following the locally stationary ARCH framework of Dahlhaus and Subba Rao (2006). Assumption 2.2 requires $\|X_\cdot\|_q < \infty$ for $q = 4$, which entails finite fourth moments of the underlying process. For a stationary ARCH(1) with constant coefficient α_1, we require $\alpha_1(u) < 1/\sqrt{3} \approx 0.577$. Likewise non-negativity implies $a - b \geq 0$. Together they define the admissible region,

$$\mathcal{V} = \big\{(a,b) :\ 0 \leq b \leq a,\ \ a + b < 1/\sqrt{3}\big\}, \tag{16}$$

To maximize the amplitude of detectable time variation while preserving the asymptotic validity of the test, we operate at the apex of $\mathcal{V}$, where both constraints in (16) bind simultaneously: $a = b_{\max} = 1/(2\sqrt{3}) \approx 0.289$. At this point the local ARCH coefficient $\alpha_1(u)$ sweeps from zero at $u = 1/2$ to the moment boundary $1/\sqrt{3}$ at $u \in \{0,1\}$, attaining the largest amplitude of time variation admissible under finite fourth moments. The null $b = 0$ is simulated alongside two alternatives $b \in \{0.14, 0.28\}$. Likewise due to the recursive nature of the process we compute the deviations after feeding a burn-in of 500 observations under the stationary regime. Results are reported in Table 3. Under the null, rejection frequencies at the 5 percent level decline monotonically with sample size at $M = 8$, from 0.127 at $T = 256$ to 0.086 at $T = 2048$, confirming asymptotic size control. Finite-sample size remains noticeably inflated at the sample sizes considered, reflecting the slow convergence of the periodogram-based variance estimator under conditional heteroskedasticity. Increasing M at fixed T raises both size and power but degrades the ratio uniformly, reaffirming $M = 8$ as the preferred choice for empirical implementation. The test retains substantial discriminatory ability against time-varying alternatives within the regime where its asymptotic theory applies, supporting its application to financial time series exhibiting ARCH-type dynamics, with the caveat that borderline rejections at finite sample sizes should be interpreted cautiously in light of the residual size inflation.

Table 3: $a = 1/(2\sqrt{3})$

			$b = 0$ (H_0)		$b = 0.14$ (H_1)		$b = 0.28$ (H_1)	
T	N	M	5%	10%	5%	10%	5%	10%
256	32	8	0.127	0.245	0.179	0.306	0.330	0.467
256	16	16	0.306	0.460	0.364	0.514	0.523	0.657
512	64	8	0.110	0.214	0.198	0.307	0.415	0.545
512	32	16	0.272	0.401	0.359	0.495	0.612	0.721
1024	128	8	0.096	0.191	0.207	0.327	0.545	0.659
1024	64	16	0.213	0.335	0.372	0.501	0.695	0.789
2048	256	8	0.086	0.167	0.238	0.350	0.709	0.804
2048	128	16	0.158	0.268	0.356	0.492	0.819	0.885
2048	64	32	0.368	0.509	0.597	0.709	0.925	0.959

4.2 Real Data

We apply the testing procedure of Section 3 to the ICE BofA US High Yield Index Option-Adjusted Spread (HY OAS, FRED series `BAMLH0A0HYM2`), a daily measure of credit risk in the U.S. high-yield bond market, observed from January 4, 2010 through January 30, 2026($T = 4014$ trading days, first differenced). The sample spans the post financial crisis zero-lower-bound era, the 2017–2018 hiking cycle, the COVID-19 shock and emergency ZLB return, the 2022–2024 hiking cycle, and the 2023 banking-stress episode. HY OAS exhibits strong conditional heteroskedasticity throughout the sample, exactly the regime in which the moment conditions of Assumption (2.2) bind tightest.

Localization

A rejection of H_0 via $\hat{D}_T^2 + B_T$ (3.5) detects that the spectrum is non-stationary somewhere over $[0, 1] \times [-\pi, \pi]$, but is unable to argue the "where." We appeal to the moving-window localization procedure of Preuss et al. (2015), hereafter PPD. Their construction supplies, at each candidate point v in rescaled time, a scalar statistic $S(v)$ that contrasts the cumulative spectral distribution of the N observations preceding v with that of the N observations following v. Like our $\hat{D}_T^2$, $S(v)$ is built from local periodograms in the sense of (8). Unlike $\hat{D}_T^2$, which integrates spectral deviation over the entire domain $[0,1]$ in u, $S(v)$ measures spectral discontinuity at the single point v. PPD additionally provide a hard threshold $\varepsilon_T(v)$, derived under their (Gaussian-linear, piecewise-stationary) framework; a candidate v is flagged as a structural break point when $S(v) > \varepsilon_T(v)$. We refer the reader to Preuss et al. (2015) for the explicit construction of $S(v)$ and $\varepsilon_T(v)$ and for the adaptive selection of the window length N. PPD's framework assumes that $f(u,\lambda)$ is piecewise constant in u, implying discrete jumps at unknown break points. Our framework $X_{t,T} = h(t/T, \mathcal{F}_t)$ admits smooth variation in u. A failure of $\hat{D}_T^2$ to reject within every PPD identified segment implies a piecewise representation, yet if we are able to identify at least one segment with a rejection, then we have empirical support to argue that the time-variation in that segment is continuous. In other words, PPD's discrete-jump form cannot capture the time-variation and motivates the smooth-u formulation of Section 2.

Testing Procedure

The testing procedure operates as follows, we first apply $\hat{D}_T^2$ to the full sample. Then, If H_0 is rejected, apply PPD's localization to extract a finite set of break points $\hat{b}_1, \ldots, \hat{b}_{\hat{K}}$. This gives us a partition of the sample at the detected breaks. On each resulting sub-segment that is long enough to satisfy Assumption 3.1, re-apply $\hat{D}_T^2$. The testing procedure then becomes recursive, if a sub-segment rejects H_0 and is large enough to support a further split, repeat the above steps within it.

In light of the $M = 8$ size results of Section 4.1, we set the minimum testable segment length to $T_{\min} = 1024$, and require a parent segment to satisfy $T \geq 2T_{\min} = 2048$ before further partitioning is attempted (so that both children remain testable).

Results

We consider the full sample, and observe that the test rejects H_0 at any conventional level: $\hat{D}_T^2 + B_T = 6.34 \times 10^{-6}$, $z = 6.85$, $p < 10^{-4}$. The series is non-stationary.

PPD's adaptive window selection over candidates $N \in \{64, 128, 256, 512, 1024\}$ returns the largest admissible $N^* = 1024$ as the detected counts $\hat{K}(N) = \{23, 18, 11, 6, 2\}$ are monotonically decreasing across the candidate range, never plateauing within the limits the sample size permits. The localization at $N^* = 1024$ produces two break points. The two detected dates align broadly with the December 2015 Fed liftoff and the January 2020 COVID emergence (Figure 1), with PPD successfully locating the two largest local discontinuities in the sample. The diagnostic of Figure 2 reveals that these are simply the two survivors of the greedy spacing rule from a much broader rejection landscape; the procedure has discretized a continuous spectral evolution into the strongest two points it can place at distance $\geq N^*$.

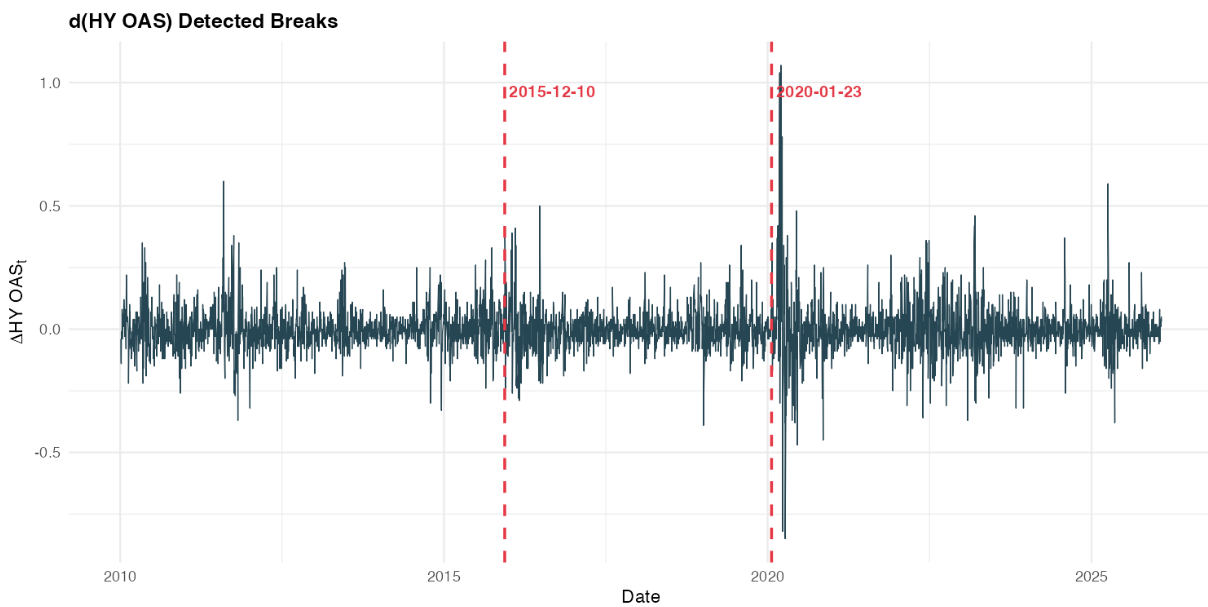

Figure 1: First differenced HY OAS over the full sample, with the two PPD-detected breaks marked

Re-applying $\hat{D}_T^2$ within each of the three resulting segments yields rejections at every level; results are reported in Table 4.

Table 4: Within-segment $\hat{D}_T^2$ results

Segment	Window	T	$\hat{D}_T^2 + B_T$	z	p
1 (ZLB)	2010-01-05 – 2015-12-10	1482	1.71×10^{-6}	2.90	0.0019
2 (Post Rate Hike)	2015-12-11 – 2020-01-23	1028	1.12×10^{-6}	2.84	0.0023
3 (post-COVID)	2020-01-24 – 2026-01-30	1503	5.53×10^{-5}	7.10	$< 10^{-4}$

All three segments fall below $2T_{\min} = 2048$ hence the procedure terminates at depth one. The piecewise decomposition implied by PPD's localization is rejected at every level of the partition: within each of the three "regimes" the spectrum continues to vary in u.

The signature of this within-regime variation is visible in the localization profile. Figure 2 plots $S(v)$ against the threshold $\varepsilon_T(v)$ at $N = N^* = 1024$ over the full sample. Pre-2016, $S(v)$ remains predominantly below $\varepsilon_T(v)$, with a single localized excursion centered on the December 2015 break. Post-2016, $S(v)$ sits broadly above $\varepsilon_T(v)$ across essentially the entire sample. Of the 1,966 candidates v at which $\varepsilon_T(v)$ is finite, only 0.6% (3 of 473) satisfy $S(v) > \varepsilon_T(v)$ pre-2016, against 98.2% (1,466 of 1,493) post-2016. Every candidate v in the post-liftoff window meets the criterion for a break candidate; PPD's greedy rule selects 2020-01-23 from this continuous rejection landscape and discards the rest, since by construction it cannot place breaks within distance N^* of one another.

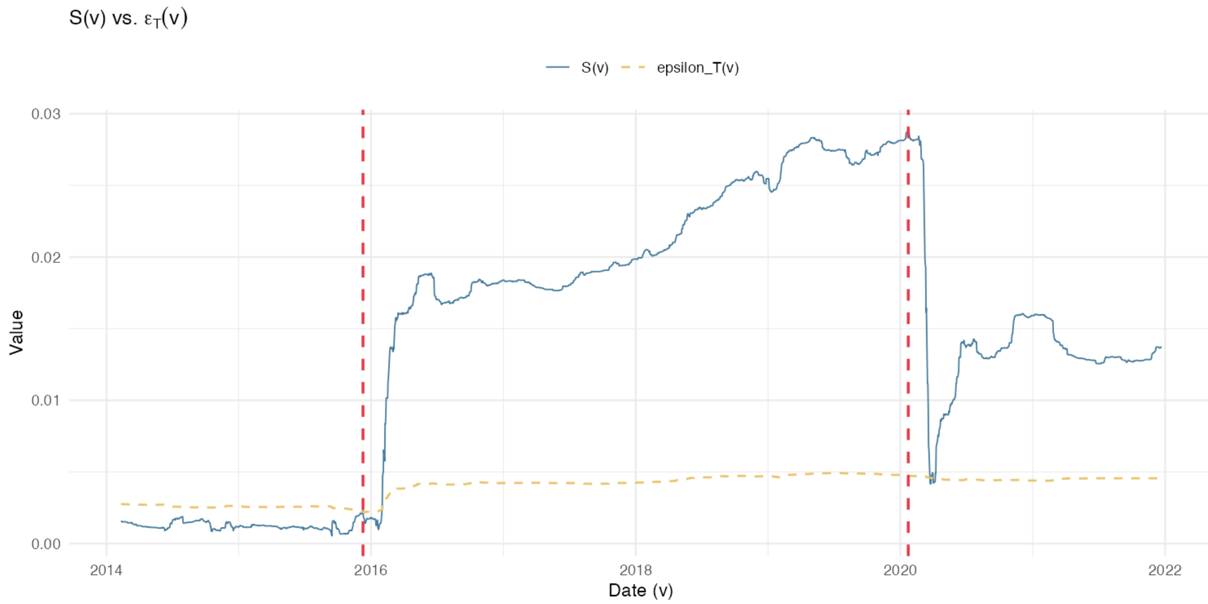

Figure 2: Localization profile $S(v)$ (solid) against the threshold $\varepsilon_T(v)$ (dashed) for first differenced HY OAS.

The post-2016 pattern is the visual signature of smooth, continuous spectral evolution rather than discrete piecewise structure. Economically, the credit market did not absorb the 2017–2018 hiking cycle, the 2020 COVID liquidity provision, the 2022–2024 hiking cycle, or the 2023 banking-stress episodes as a single discrete shift; their cumulative effect on the conditional volatility of HY OAS appears as continuous time-variation in $f(u, \lambda)$. PPD's piecewise representation has no mechanism to absorb such variation. The framework $X_{t,T} = h(t/T, \mathcal{F}_t)$ of Section 2 does, and the empirical evidence above supports its adoption for this series.

References

R. Dahlhaus. Fitting time series models to nonstationary processes. *The Annals of Statistics*, 25(1):1–37, 02 1997. 2

R. Dahlhaus and W. Polonik. Empirical spectral processes for locally stationary time series. *Bernoulli*, 15(1):1–39, 02 2009. 2

R. Dahlhaus and S. Subba Rao. Statistical inference for time-varying arch processes. *The Annals of Statistics*, 34(3):1075–1114, 2006. 2, 14

R. Dahlhaus and S. Subba Rao. A recursive online algorithm for the estimation of time-varying arch parameters. *Bernoulli*, 13(2):389–422, 2007. 2

Holger Dette and Efstathios Paparoditis. Bootstrapping frequency domain tests in multivariate time series with an application to comparing spectral densities. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 71(4):831–857, 2009. 1

Holger Dette, Philip Preuss, and Mathias Vetter. A measure of stationarity in locally stationary processes with applications to testing. *Journal of the American Statistical Association*, 106(495):1113–1124, 2011. 1, 4

Yogesh Dwivedi and Suhasini Subba Rao. A test for second-order stationarity of a time series based on the discrete fourier transform. *Journal of Time Series Analysis*, 32(1): 68–91, 2011. 1

Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007, 1982. 14

P. Fryzlewicz and S. Subba Rao. Mixing properties of arch and time-varying arch processes. *Bernoulli*, 17(1):320–346, 2011. 2

Y. Grenier. Time-dependent arma modeling of nonstationary signals. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 31(4):899–911, 1983. 2

C.M. Hafner and O. Linton. Efficient estimation of a multivariate multiplicative volatility model. *Journal of Econometrics*, 159(1):55–73, 2010. 2

E. Moulines, P. Priouret, and F. Roueff. On recursive estimation for time varying autoregressive processes. *The Annals of statistics*, 33(6):2610–2654, 2005. 2

Philip Preuss, Ruprecht Puchstein, and Holger Dette. Detection of multiple structural breaks in multivariate time series. *Journal of the American Statistical Association*, 110 (510):654–668, 2015. 15, 16

M. B. Priestley and T. Subba Rao. A test for non-stationarity of time-series. *Journal of the Royal Statistical Society. Series B (Methodological)*, 31(1):140–149, 1969. ISSN 00359246. 1

M.B. Priestley. *Nonlinear and Nonstationary Time Series Analysis.* Academic Press, London, 1988. 3

M. Rosenblatt. *Markov Processes: Structure and Asymptotic Behavior.* Springer, 1971. 3

X. Shao and W.B. Wu. Asymptotic spectral theory for nonlinear time series. *The Annals of Statistics*, 35(4):1773–1801, 2007. 5

T. Subba Rao. The fitting of non-stationary time-series models with time-dependent parameters. *Journal of the Royal Statistical Society. Series B (Methodological)*, 32(2): 312–322, 1970. 2

H. Tong. *Non-linear Time Series: A Dynamical System Approach.* Oxford University Press, 1990. 3

R.S. Tsay. *Analysis of Financial Time Series*, volume 543. John Wiley & Sons, 2005. 3

N. Wiener. *Nonlinear Problems in Random Theory.* Wiley, New York, 1958. 3

W.B. Wu. Nonlinear system theory: another look at dependence. *Proceedings of the National Academy of Sciences of the United States of America*, 102(40):14150–14154, 2005. 3

H. Xiao and W. B. Wu. Asymptotic inference of autocovariances of stationary processes. *arXiv preprint arXiv:1105.3423*, 2011. 10

Danna Zhang and Wei Biao Wu. Convergence of covariance and spectral density estimates for high-dimensional locally stationary processes. *The Annals of Statistics*, 49(1):233–254, 2021. 5

Z. Zhou. Nonparametric inference of quantile curves for nonstationary time series. *The Annals of Statistics*, 38(4):2187–2217, 08 2010. 2